

Machine learning methods for continuous glucose monitoring data and their potential in cardiovascular risk assessment

PhD dissertation

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1 Introduction

1.1. Diabetes and cardiovascular disease

- Burden of diabetes
 - Type 1 diabetes
 - Type 2 diabetes
- Cardiovascular complications
- Heart failure in diabetes
- NT-proBNP as a biomarker

1.2. Glycemic markers and glucose monitoring

- HbA1c and its limitations
- Continuous glucose monitoring (CGM)
- Glycemic variability
 - Metrics of glycemic variability
 - Clinical challenges of glycemic variability (Think HbA1c and MBG and variability study)

1.3. Cardiovascular risk prediction in diabetes

- Traditional risk models
 - SCORE2-Diabetes risk score
 - What is SCORE2-Diabetes 10 year CVD risk score?
 - ☐ What is the SCORE2-diabetes and why was it chosen?
- Limitations of conventional predictors

1.4. Machine learning

- Data-driven vs hypothesis-driven(knowledge-based) paradigms
- Why machine learning in diabetes and cardiovascular risk prediction?
- Why are ML fantastic for CGM data?
- Types of machine learning methods
 - Logistic regression
 - LSTM and CNNs

- Foundation models
 - * CGM data has been processed and interpreted based on plotting the data and extracting CGM-derived metrics from time series signal sampled every 3-15 minutes. However new techniques within the machine learning field has allowed us to process the whole time series signals like CGM data, which can retain more information from the original CGM data then the derived metrics can. Foundation models are machine learning models that are trained on huge amount of data and thereafter can be retrained to be specialized into a specific task. This is one solution to the limitation of small datasets that are not sufficient enough for the data hungry machine learning models.
 - * The CGMformer is a transformer-based model that was trained on a large dataset of CGM data and is designed to learn representations of glucose dynamics. The model takes in sequences of CGM measurements and outputs embeddings that capture the underlying patterns in the glucose data. These embeddings can then be used for downstream tasks such as predicting cardiovascular risk or other health outcomes.
 - * The embeddings are learned representations that capture the underlying patterns and relationships in the input data, allowing for more efficient and effective downstream tasks such as classification, regression, or clustering.

The CGMformer is bla bla bla¹ * Challenges and pitfalls of machine learning in healthcare * Overfitting and generalizability * Interpretability and explainability

1.5. Fairness and bias in machine learning

- Define race and ethnicity
- Algorithmic bias and group fairness
- Fairness metrics and mitigation strategies
- Challenges in healthcare applications when using ML

2 Aims

2.1. Overall aims

The overall aim of this PhD dissertation was to investigate the potential of continuous glucose monitoring (CGM) data to improve cardiovascular risk assessment and to evaluate the role of machine learning, including foundation model-derived glucose representations, in extracting clinically relevant information from CGM data.

2.2. Specific aims

- I. To map the existing literature on the association between glycemic control and cardiovascular outcomes and identify CGM-derived metrics investigated for cardiovascular risk assessment.
- II. To determine whether a CGM foundation model captures meaningful and interpretable glucose representations across individuals with type 1 and type 2 diabetes.
- III. To evaluate whether conventional CGM-derived metrics and learned glucose representations improve prediction of elevated NT-proBNP beyond established clinical risk factors in individuals with type 2 diabetes.”
- IV. To evaluate the fairness of machine learning models for glucose prediction and compare strategies for tailoring prediction models to individual characteristics.

Methods 3

In this chapter the three studies included in this PhD will be presented. The data will be described together with the chosen methods of the three studies and the different clinical variables, CGM-derived metrics and relevant cardiovascular outcomes will be explained.

- ☐ What was done (Description)
- ☐ Why it was done (Justification)
- ☐ Add search string?
- ☐ tell about the 7-point fingerprick?

3.1. Overview of the studies

	Study I	Study II	Study III
Title	Continuous Glucose Monitoring-Derived Metrics and Cardiovascular Risk Among People With Diabetes: Systematic Scoping Review	Foundation model-derived representations of continuous glucose monitoring data: relation to glycemic patterns and cardiovascular risk Racial disparities in continuous glucose monitoring-based 60-min glucose predictions among people with type 1 diabetes	
Study Type	Review	CGM foundation model evaluation study	Algorithmic fairness study
Data Sources	Literature on Embase and MEDLINE	The CANCAN study and the T1D Exchange (T1DX) dataset	T1D Exchange (T1DX) dataset
Population	Studies including people with type 1 diabetes, type 2 diabetes and/or prediabetes	159 people with type 2 diabetes 205 people with type 1 Diabetes	205 people with type 1 Diabetes
Input, predictors and exposures	Search strategy and eligibility criteria	Clinical variables, CGM-derived metrics and continuous CGM data	Continuous CGM data
Outcome, target	All studies that uses CGM data to say something about cardiovascular disease risk and outcomes	Visualisation of foundation model embeddings and elevated NT-proBNP prediction	Next blood glucose value

	Study I	Study II	Study III
Analysis	Systematic scoping review, study characterization, and narrative synthesis	Foundation models and logistic regression models	Long short term memory models (LSTM) and Linear mixed effect models (LMEM)

3.2. Study I: A scoping Review

□ Explain why scoping review and not a systematic review

A scoping review was conducted to investigate current knowledge of CGM-data and its possible connection and predictive power to clinical and subclinical cardiovascular outcomes. Given the expected heterogeneity between studies, a scoping review was considered the most appropriate methodology. The scoping review followed the Manual for Evidence Synthesis (Chapter 10 - Scoping reviews) from the Joanna Briggs Institute and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses extension for scoping review (PRISMA-ScR) guidelines.^{2,3}

3.2.1. Concepts and Definitions

The review focused solely on CGM-data for the input to any association or prediction models in the literature. This meant that studies only investigating self-monitored blood glucose with finger-prick measurements or blood sample measurements such as HbA1c were not considered in this review. CVD outcomes were divided into subclinical or clinical For the clinical CVD, the following outcomes were considered: cardiovascular mortality, major adverse cardiovascular events, coronary artery disease, heart failure, stroke, and peripheral artery disease with synonymous terms included (e.g., ischemic heart diseases) and also clinical events (e.g., undergoing coronary artery bypass surgery). Subclinical CVD was grouped into six subcategories: arterial stiffness, flow resistance, arterial wall thickness, arterial wall composition, cardiac and pulse-related measures, and arterial lumen. CVD outcomes did not include broader risk factors nonspecific to cardiovascular risk (e.g., age or sex). Eligibility criterias for the scoping review⁴ (Study I) can be found in Table 3.1.

Table 3.1.: Eligibility criteria for the scoping review (Study I)

Inclusion Criteria	Exclusion Criteria
Human clinical studies including participants with prediabetes or any type of diabetes except for gestational diabetes, regardless of definitions	Review articles, editorials, case reports, protocols, conference abstracts, and preprints.
Peer-reviewed published original articles (including brief reports).	Animal studies not including any human participants.
Studies investigating either:	Studies not including metrics derived from continuous glucose monitoring device data.
a) The association between CGM-derived metrics of glycemic control and cardiovascular risk markers or cardiovascular diseases (prevalent or incident).	Studies including CGM-derived metrics as outcomes.
b) CGM-derived metrics of glycemic control as predictors of cardiovascular risk markers or cardiovascular diseases (prevalent or incident).	Studies not focusing on CVD outcomes according to our definition, as outlined in the “Concepts and Definitions” section.
	Studies focusing on pregnant women with any form of diabetes including gestational diabetes.
	Studies where participants were monitored post-surgery or during hospitalization (e.g. Intensive care unit).
	Language not understood by authors

3.2.2. Search and Selection of Sources

The databases Embase and Medline were searched from inception to March 11, 2025. The search strategy was tested against 13 articles in the field, found through an initial exploratory search.^{5–17}. The full search strategy can be found in Appendix A.

All identified records were uploaded into EPPI reviewer 6¹⁸ and duplicates were removed. Titles and abstracts were screened by at least two independent reviewers and a meeting was held after ~5% of the titles and abstracts had been screened, to resolve disagreements and ensure consistency between reviewers. Afterwards full-text screening was performed by two independent reviewers and held up against eligibility criteria. Disagreements were resolved through discussion at physical meetings between reviewers.

Backward and forward citation searching was performed on all the included studies after the screening process using citationchaser¹⁹ and the screen process was repeated until no further new studies were identified through the citationchaser.

3.2.3. Data Charting Process and Data Items

Research questions were defined before the screening process and data was extracted based on these research questions (Note 1) and following the requirements from the PRISMA-ScR checklist.²

Note 1: Research Questions (Study 1)

1. Is there an association between glycemic control and CVD risk?
2. Can CGM-derived metrics predict CVD risk?
3. What CGM-derived metrics are used in the literature?

4. Which cardiovascular markers and diseases are included as outcomes in the studies?
5. What characterizes study populations (age, sex, ethnicity, geographic location)?
6. What study designs are used (e.g., longitudinal cohort studies, randomized controlled trials, cross-sectional studies)?
7. How was data collected (e.g., clinical trial, epidemiological study, routinely collected data)?
8. What CGM devices were used?
9. What statistical models were used in the studies?
10. Is the data openly available?
11. Is the code openly available?

3.2.4. Synthesis of Results

Study characteristics were synthesized using descriptive statistics and supplemented by a narrative summary. Only the most adjusted models in the main population for each study was considered in this review. This approach was necessary as several studies reported numerous estimates across varying adjustments, different combinations of CGM-derived metrics and CVD outcomes, and/or distinct subgroup stratifications. Given the large volume of literature, studies investigating clinical cardiovascular outcomes were prioritized in the main synthesis. While subclinical markers offer valuable insight into early vascular changes and disease progression, clinical outcomes are more directly relevant to cardiovascular risk prediction and clinical decision-making. The main synthesis of clinical outcomes included both effect estimates and measures of statistical significance whereas the supplementary synthesis on subclinical CVD focused primarily on creating an overview of the chosen CGM-derived metrics and the statistical significance of reported associations. All evaluation metrics mentioned in the studies doing prediction analyses were reported based on what the studies reported.

3.3. Study Populations

Two datasets were used in this PhD (Study II and study III) which are presented in the following sections

3.3.1. CANCAN

The CANCAN dataset was collected in an observational case-control study from three hospital outpatient clinics (Denmark). The aim of the CANCAN Study was to investigate if cardiac autonomic neuropathy (CAN) assessments, CGM and heart failure indicators could be used in risk stratification in people with T2D. Participants were included if they were adults (< 18 years old) and had T2D for at least a year. Exclusion criteria were atrial fibrillation or laser-treated eye disease within the past three months, were pregnant or lactating, had a life-threatening disease (expected survival < 1 year), or had cognitive disabilities rendering them unable to understand and give consent. Data were collected during a clinical examination, during which blood and urine samples were obtained, and a blinded CGM sensor was inserted. Blood samples to obtain NT-proBNP values were collected when the CGM sensors were returned, after the 2-week CGM data collection period had ended (Figure 3.1). In total, 159 participants were included since the rest were missing age, sex, HbA1c, BMI, NT-proBNP, HDL, total cholesterol, or eGFR and these values were needed for the analyses in Study II.

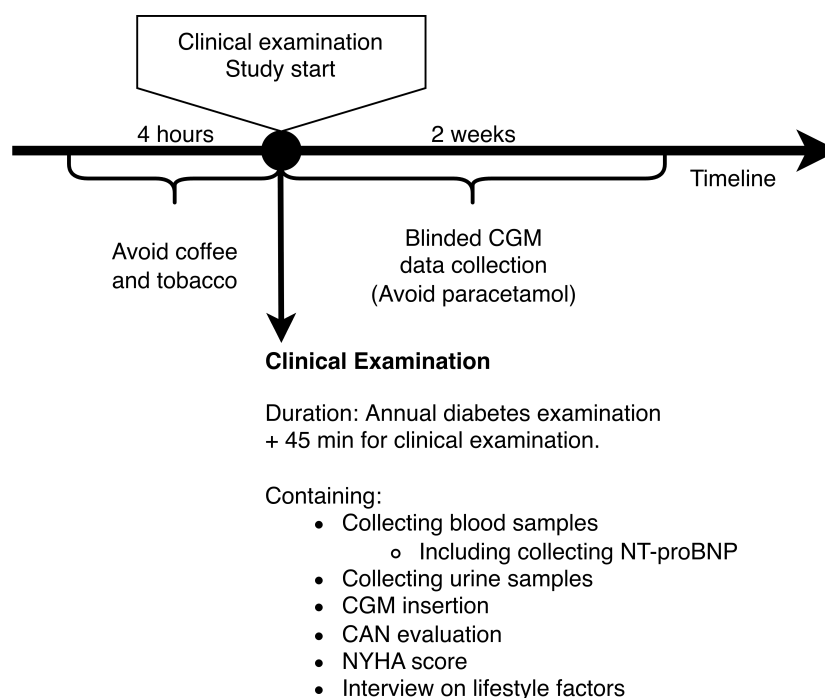


Figure 3.1.: Data collection timeline for the CANCAN Study

3.3.2. T1DX

The publicly available T1D Exchange (T1DX) dataset, distributed by the JAEB Center for Health Research under the protocol “Racial Differences in Mean CGM Glucose in Relation to HbA1c, was collected in 10 diabetes centers (US).²⁰ The data was collected to determine if there is a racial difference in the association between mean glucose and glycated hemoglobin (HbA1c) between non-Hispanic African Americans or black (Black) and non-Hispanic white (White) people with T1D. In the T1DX study ethnicity and race was defined through self-identified and self-reported ethnicity/race verification forms where ethnicity was defined as Hispanic or Latino as: “a person of Cuban, Mexican, Puerto Rican, South or Central American, or other Spanish culture or origin, regardless of race. The term “Spanish origin” could be used in addition to “Hispanic” or “Latino””.²¹ For race a participant defined as White was “a person having origins in any of the original peoples of Europe, the Middle East or North Africa.” and Black or African American as “a person having origins in any of the Black racial groups of Africa. Terms such as “Haitian” or “Negro” can be used in addition to “Black” or “African American””.²¹ In this thesis, the term race will be used moving forward, as it aligns with the terminology used by Bergenstal et al.²¹, it is, however, acknowledge that it is inherently difficult to define race.²² The T1DX study adhered to the tenets of the Declaration of Helsinki and was approved by the local institutional review boards and 205 people completed the study by providing written informed consent and assent when applicable.²¹ All 205 participants were included in further analyses. Of the 205 participants, 101 were White and 104 were Black. For each race, two age groups were sampled 8 to <18 years old and >18. Each participant wore a blinded modified investigational version of Abbott Diabetes Care’s Flash Glucose Monitoring System (CGM) as continuously as possible for 12 weeks. CGM data was sampled every 15 minutes in a range between 2.21 - 27.65 mmol/L (40 and 498 mg/dL). Blood samples were taken for measurement at enrollment, at some of the follow-up visits, and at the end of enrollment.

3.4. Study II: Foundation models and predictive utility of CGM data

The second study was focused on investigating the knowledge and predictive utility of CGM data, both extracting a novel CGM-derived metric and the classical known ones found in study I.⁴

3.4.1. Data and Participant Selection

Both the CANCAN and the T1DX dataset was used for Study II. Participants with missing age, sex, HbA1c or BMI were removed for both cohorts and additional CANCAN participants with missing NT-proBNP, HDL, total cholesterol, and eGFR were also removed, since it was wished to calculate the SCORE2-Diabetes risk score. T1DX participants with missing race data were also removed. Days with less than 70% of CGM data were removed as well as participants with less than one day of data from both datasets to ensure sufficient data for the analyses. The final number of participants included in Study II was 159 from the CANCAN dataset and 205 from the T1DX dataset.

3.4.2. Embeddings

Two promising CGM specific foundation models exist, both published within the last year.²³ Of the two foundation models, only one model is fully open access, the CGMformer.¹ The CGMformer was therefore chosen and the data was organized in 24-hour long-format input to match the input requirements of the CGMformer. The output of the foundation model is an embedding, which is a numerical vector that summarizes the characteristics of the input glucose profile. A simple vector can be drawn as an arrow in a 2-dimensional space as shown in Figure 3.2, however the CGMformer gives a 128-dimensional embedding vector for each 24-hour CGM data input which is not possible to visualize. Uniform Manifold Approximation and Projection (UMAP) algorithm was used to reduce the dimensionality of the embeddings to two dimensions for visualization purposes and to compare with current literature.^{1,23,24} The UMAPs were colored by TIR, and MAGE to investigate if the embeddings were able to capture relevant information about glycemic patterns. History of CVD and elevated NT-proBNP for the CANCAN embeddings were also used to color the UMAPs to investigate if the embeddings were able to capture relevant information about cardiovascular risk and race was used to color the T1DX embeddings to investigate if the embeddings were able to capture relevant information about racial differences in glycemic patterns.

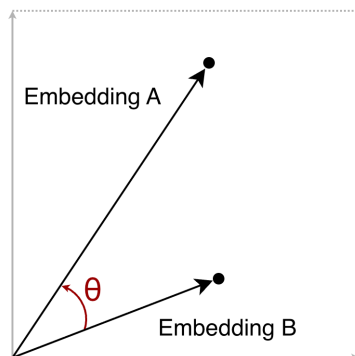


Figure 3.2.: Embeddings shown as 2-dimensional vectors with the angle θ between the two embeddings.

Because embeddings are numerical vectors, they can be compared to quantify the similarity between glucose profiles. Cosine similarity was used because it measures the angle between

two embedding vectors independent of their magnitude (the length of the vectors). The cosine similarity is calculated by measuring the cosine angle between two vectors as shown by θ in Figure 3.2. The cosine similarity is a number between -1 and 1, where 1 means the two vectors are identical, 0 means they are unrelated, and -1 means that they are pointing in opposite directions (Figure 3.3). For the analyses in this study, cosine distance ($1 - \text{cosine similarity}$) was used because it expresses similarity as a distance measure, allowing embeddings with similar orientations to have small distances and facilitating downstream analyses and visualizations. The cosine distances range from 0 to 2 where 0 means the two vectors are identical, 1 means they are unrelated, and 2 means that they are pointing in opposite directions.

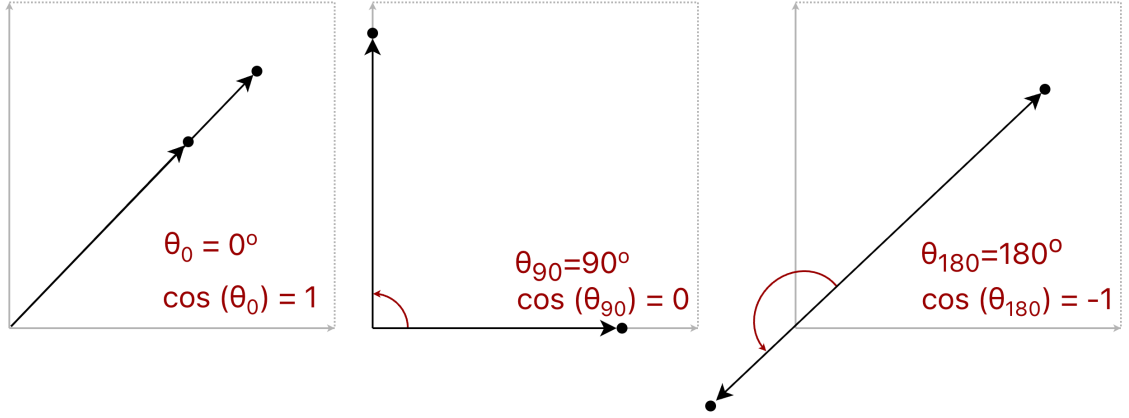


Figure 3.3.: Embeddings shown with different cosine similarities. The cosine similarity is calculated as the cosine of the angle (θ) between two embeddings.

Kernel density plots were used to visualize the distribution of cosine distances within specific clinical strata and across differing participants (Figure 3.4). Groupings were made based on days from the same participant, days from different participants with the same clinical strata, and days from different participants with different clinical strata. The main focus was on TIR and HbA1c were the thresholds were as followed: TIR 70% (70–180 mg/dL, 3.9–10.0 mmol/L) and HbA1c <7.0 % (< 53 mmol/mol). Other UMAPS coloring and groupings were also explored including race, MAGE, previous history of CVD, NT-proBNP, and SCORE2-Diabetes scores.

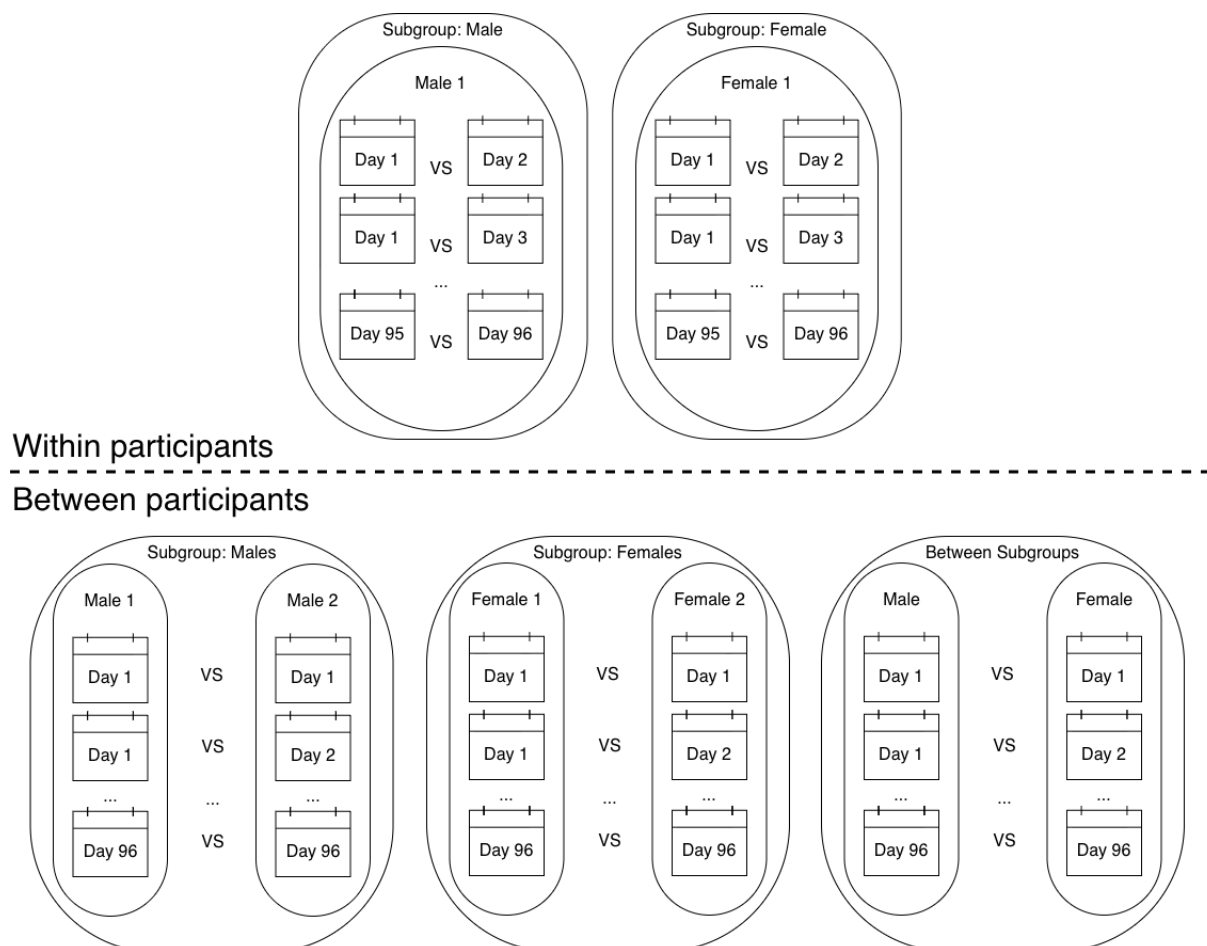


Figure 3.4.: The sex strata is used as an example to illustrate the cosine distance analysis of the within and between cosine distance pairs

3.5. Cardiovascular risk assessment

The following section is about cardiovascular risk assessment evaluated predicting NT-proBNP as a binary outcome using a >125 pg/mL threshold.^{source?} The CANCAN dataset was the only dataset used for this CVD risk assessment. The focus was to test the predictive power of CGM data both as represented as embeddings and as well-established CGM-derived metrics. Some models also included clinical predictors to compare up against baseline models.^{source_kovatchev?orAnotherSourceOnMAGE?} Prediction models integrated clinical, standard CGM-derived, and learned representations.

3.5.1. NT-proBNP - The target

- ☐ What is NT-proBNP (scale, unit and connection with heart failure)

3.5.2. Predictors

- ☐ Add table with CGM derived metrics?

Inspiration was drawn from the SCORE2-Diabetes risk score, which is a 10-year CVD risk score for European people with diabetes. It was decided to use the input predictors of the SCORE2-Diabetes risk score as the clinical set of predictors used in this study.^{source?} This resulted in a set of clinical predictors including age, sex, smoking status, systolic blood pressure, total cholesterol, and HbA1c.^{source?} Initial testing quickly showed that the numbers of predictors had to be limited and the SCORE2-Diabetes risk score was chosen as another way to represent the clinical predictor set while keeping the number of predictors down. Additionally age was tested as a predictor on its own to see if it was a strong predictor of elevated NT-proBNP, since NT-proBNP is known to increase with age.²⁶ Time in range (TIR) and mean amplitude of glycemic excursions (MAGE) were chosen as they were the CGM-derived metrics most commonly used in the literature based on the findings from Study I.⁴ All CGM-derived metrics in this thesis were calculated using the IGLU package^{source?} in R with default settings. TIR was calculated as the percentage of time spent in the range of 3.9–10.0 mmol/L (70–180 mg/dL).²⁵ MAGE was calculated as the mean amplitude of glucose excursions, where the included excursions were consecutive peak-to-nadir (or nadir-to-peak) glucose differences exceeding one standard deviation of the individual's glucose measurements.^{Does} The iglu implementation computationally reproduces the original manual procedure proposed by Service et al.²⁷ To aggregate multiple days into a single observation per participant, daily CGMformer representations were minimum-, maximum-, and mean-pooled, and reduced to two dimensions via UMAP. Maximum pooling was considered the default based on previous work.²³ The CGMformer generated one 128-dimensional embedding for each 24-hour CGM recording. Because NT-proBNP was measured at the participant level, a single embedding was required for each participant. Therefore, embeddings from multiple recording days were aggregated using pooling to obtain one participant-level embedding. Both minimum, maximum, and mean pooling were explored to determine the most effective aggregation method. The resulting 128-dimensional embeddings were subsequently reduced to two dimensions using UMAP, thereby enabling their use in the prediction models while keeping the number of predictors down to a minimum.

3.5.3. Model development and evaluation

- ☐ Check if any of the models pulled from supplementary uses ridge and if yes, then include it here as well and explain why ridge is used

Prediction models were developed using logistic regression to predict elevated NT-proBNP (>125 pg/mL) as a binary outcome. The models were evaluated using the area under the receiver operating characteristic curve (AUROC) as the primary metric. Bootstrap with optimism correction was employed for internal validation because of the limited sample size and the absence of an external validation cohort. The method repeatedly sampled bootstrap samples by sampling with replacement from the original dataset. For each bootstrap sample, the prediction model is fitted and its performance is evaluated both in the bootstrap sample (apparent performance) and in the original dataset (test performance). The difference between these two performance estimates represents the optimism due to overfitting. Averaging the optimism across all bootstrap samples and subtracting it from the apparent performance of the model fitted to the full dataset yields an optimism-corrected estimate of model performance. Internal validation was performed using 500 bootstrap resamples.^{28,29} Risk distribution plots were reported for each model across all bootstrap samples.^{29,30}

3.5.4. Prediction scheme overview

The prediction models were structured hierarchically based on clinical variables (Model A) and the calculated SCORE2-Diabetes score (Model B) in the configurations shown in `?@tbl-models`

Table 3.2.: Overview of models and predictors {#tbl-models}Grr

Model family	Model	Baseline predictors	Additional predictors
Clinical	A0	Age	–
Clinical	A1	Clinical variables*	–
Clinical	A2	Clinical variables*	TIR + MAGE
Clinical	A3	Clinical variables*	UMAP dimensions
SCORE2-Diabetes	B1	SCORE2-Diabetes score	–
SCORE2-Diabetes	B2	SCORE2-Diabetes score	TIR + MAGE
SCORE2-Diabetes	B3	SCORE2-Diabetes score	UMAP dimensions
CGM only	C1	TIR + MAGE	–
Embeddings only	D1	UMAP dimensions	

* *Clinical variables are the same as those included in the SCORE2-Diabetes risk model: Age, sex, smoking status, systolic blood pressure, HDL cholesterol, total cholesterol, diabetes duration, HbA1c, and eGFR.*³¹

TIR = Time in Range; MAGE = Mean Amplitude of Glycemic Excursions.

3.6. Study III: Algorithmic Fairness in Blood Glucose Forecasting

The third study was investigating racial bias in a data driven machine learning model only based on blood glucose measurements from a CGM-sensor as this is an important consideration in the path to get ML models into use

3.6.1. Preprocessing of CGM data for Forecasting

All data was sorted into 60 minutes windows with 4 values to predict 60-minutes into the future as visualized in Figure 3.5. There are no clear answers in the literature about which prediction horizon is the best to choose, however CGM devices measure in the interstitial fluid and there is a physiological delay between the measured blood glucose and the interstitial glucose levels.^{source?} Of the European approved CGM devices it is only the Dexcom that shows trends to the users and since CGM devices mainly focuses on monitoring, the forecasting question is only really interesting when administering insulin in order to catch hypo- and hyperglycemic events ahead of time.^{source?} It is very difficult to figure out how the prediction algorithms in insulin pumps are made and so far I have had very little luck in figuring it out. It is assumed that the companies use linear regression and the last 20 minutes of data to predict the next values which is usually the last four samples since the resolution of the used CGM devices usually are between 1 to 5 minutes.^{source?} With a CGM measurement every 15 minutes and the physiological delay in mind, the 60 minutes were decided to be a good prediction horizon. All samples with missing values were removed.

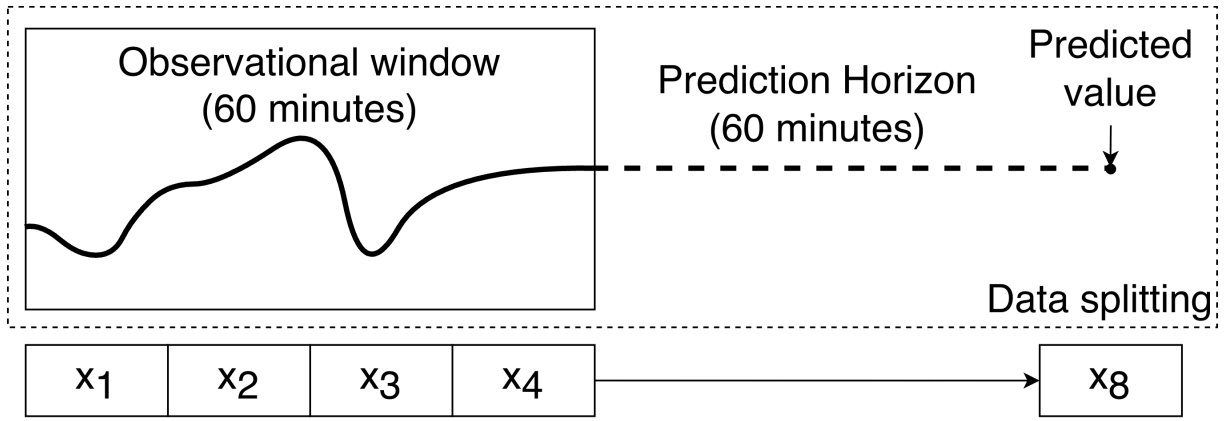


Figure 3.5.: The prediction horizon and overview of the creation of samples (predictors and targets).

3.6.2. Partitioning of the Dataset

The T1DX dataset was randomly resampled without replacement at the participant level to avoid data leakage and to evaluate model performance on independent participants. As illustrated in Figure 3.6, training subsets with varying proportions of Black and White participants were generated by randomly selecting participants from the original dataset while keeping the train-test split at the participant level. Eleven training set compositions were created, ranging from 100% White/0% Black to 0% White/100% Black in 10% increments.

Because some training subsets consisted entirely of participants from a single racial group, the maximum training set size was constrained by the size of the smaller racial group minus the participant kept for testing. Consequently, each training subset contained approximately half the number of participants in the original T1DX dataset. Repeating this procedure for each participant resulted in 2,255 training datasets ($205 \text{ participants} \times 11 \text{ training set compositions}$) and 205 independent test sets. Throughout all resampling procedures, all CGM recordings from a participant were kept together to prevent data leakage between the training and test sets.

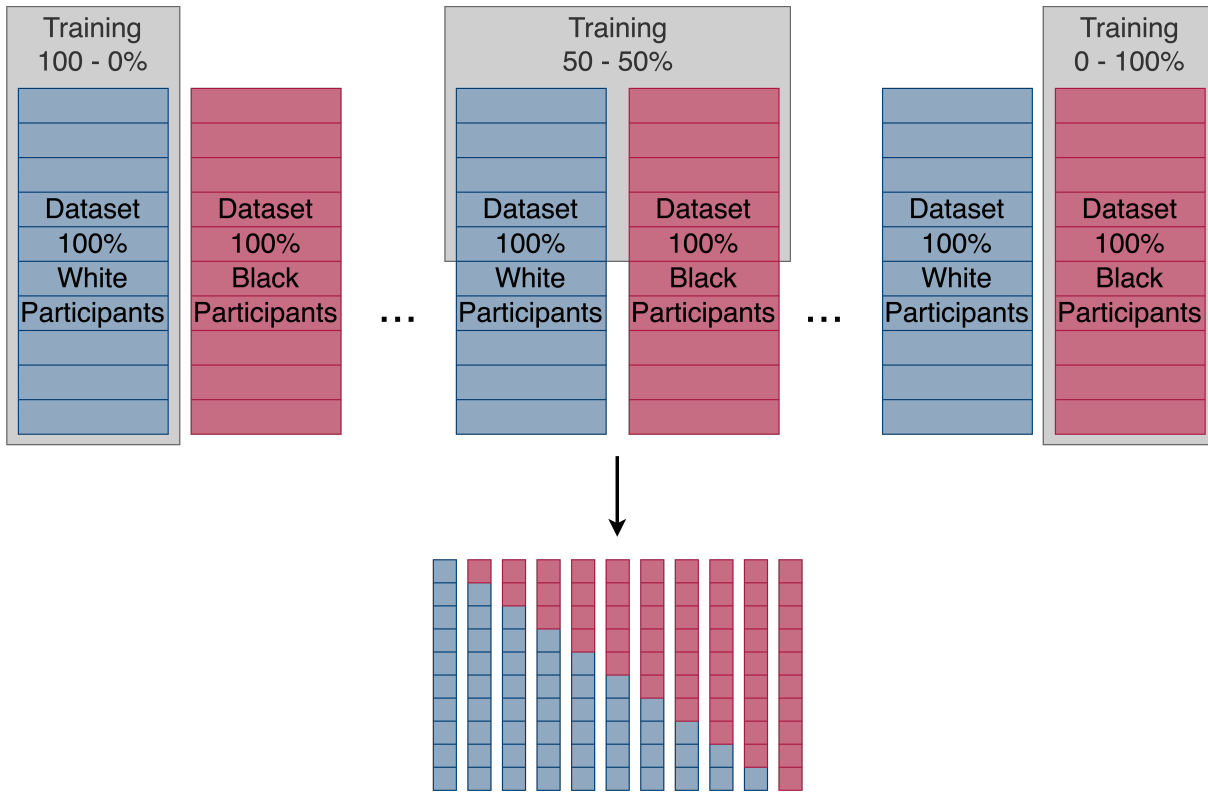


Figure 3.6.: The prediction horizon and overview of the creation of subsets for training.

Eleven subsets were created for each of the participants and the last weeks worth of data was chosen to test on as illustrated in Figure 3.7. The one week was chosen because participants had varying amount of CGM data and the participant with the least amount of data had ~ 2 weeks, and all participants were wished to be kept due to the small number of participants in the T1DX dataset. The remaining data were used for training the models.

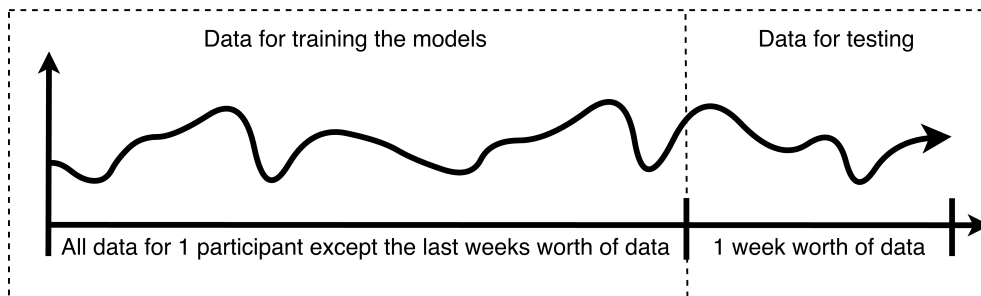


Figure 3.7.: The splitting of a participants data to create training, fine-tuning and test sets

3.6.3. Training Approaches and Model Selection

Four training approaches were developed as illustrated in Figure 3.8. The training approaches included a naive baseline and three different machine learning models as described in Table 3.3.

A long short-term memory model (LSTM) and overall architecture was chosen for the machine learning model, based on a previously described LSTM in van Doorn et al. @vanDoorn2021 (two LSTM layers followed by a dense layer) since there were good performance with this model. All the training data were scaled to be within the range of zero to one. Hyperparameters included the number of neurons in LSTM layers, activation functions, dropout rate, batch size, learning rate, decay rate, and early stopping. These hyperparameters were tuned using a grid search,

heatmaps, and manual tuning. Further information is given in in Thomsen et al.³² and the code repository³³.

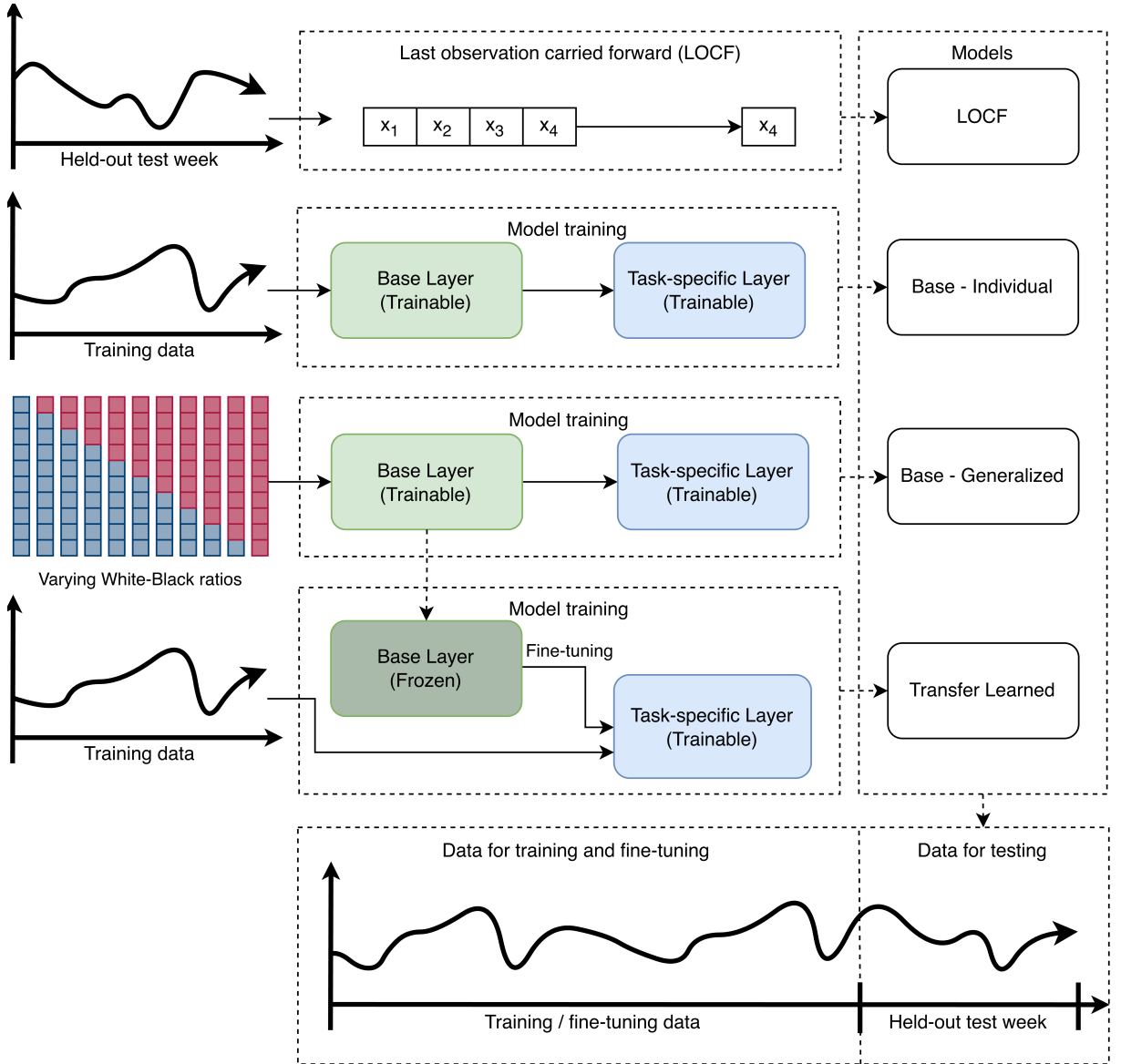


Figure 3.8.: The four training approaches. The Base-Individual, Base-Generalized, and Transfer Learned models were based on an LSTM architecture. The base layer (green) consisted of two LSTM layers, while the task-specific layer (blue) consisted of a single dense layer.

One of the machine learning models were developed using transfer learning. It is a machine learning technique in which knowledge learned from one domain is transferred to another related domain. One common transfer learning approach is fine-tuning, where a pretrained model is further trained using new data.³⁴ In this analysis, generalized glucose prediction models were first trained using data from multiple participants (Base-Generalized model) and subsequently fine-tuned using participant-specific data to personalize glucose predictions (Transfer Learned model).

Table 3.3.: The four training approaches used to develop glucose prediction models for each participant in the T1DX dataset.

Model names	Description	Training data
Last observation carried forward (LOCF)	naive baseline: In the LOCF model, a given CGM measurement (x4) was used to predict blood glucose values 60 minutes in advance (x8) in the test set by assuming no change in the last 60 minutes.	No training data required
Base-Individual	In the Base-Individual model, each participant's complete CGM data (except for the last week, which was used for the testing data) was used for training.	Only data from 1 participant Figure 3.7
Base-Generalized	Here, for each participant, the 11 training corresponding datasets described above were used to generate 11 separate models, which were all subsequently tested on the given participant's last week of CGM data.	All the different subsets of data Figure 3.6
Transfer Learned	This model mirrored the Base-Generalized Model's setup. However, the 11 trained models for each participant were subsequently tailored, i.e., fine-tuned with transfer learning to the given participant's complete CGM data (except for the last week, which was used for the testing data)	Only data from 1 participant Figure 3.7

3.6.4. Model Evaluation

To evaluate model performance, each participant's last weeks worth of data (672 measurements) was reserved for internal testing (Figure 3.7). Model accuracy was assessed by calculating the root mean squared error (RMSE) for each participant across all four modeling approaches. For the LOCF and Base-Individual models, this yielded one RMSE value for each of the 101 White participants and 104 Black participants. For the Base-Generalized and Transfer Learned models, RMSEs were calculated for each participant across 11 data subsets, each representing a unique proportion of White and Black participants. Performance was summarized as the mean RMSE with corresponding 95% confidence intervals (CIs) for each racial group.

To evaluate the predictive performance across different ratios for the Base-Generalized and Transfer Learned models a linear mixed effect model (LMEM) was used since data was dependent due to participants having an RMSE value for each subset with varying proportions of White and Black participants. It was assumed that RMSE values would vary between individuals and therefore a random intercept was included and the following covariates were included: ratio (proportion of White and Black), race, age, sex, and training data sample size. (Categorical: Race, age, and sex; continuous: RMSE, ratio, and sample size) It was further assumed that the varying proportions of Black and White participants in the training data subsets would affect the performance and an interaction term was therefore added. The interaction term allowed for different slopes in the LMEM which allowed us to test the hypothesis

Surveillance error grids (SEG) were used to get a general idea of the clinical risk of the prediction models.³⁵ The error grids are originally developed for the regulatory bodies to assess clinical accuracy of blood glucose monitoring in post-surveillance before the wide spread use of CGM devices.³⁶ and the results should therefore be interpreted with caution when using CGM data. The SEG is a 600x600 grid assigned an individual risk score to each datapoint with 15 colored zones, each indicating severity of a monitoring error that is acted upon in regards to risk of hypoglycemia or hyperglycemia (Figure 3.9).³⁶ To facilitate interpretation, the 15 surveillance error grid zones were collapsed into the five broader risk zones ("No", "Slight", "Moderate",

“High”, and “Extreme”), as previously suggested in the literature.³⁵ Predictions outside of 0–33 mmol/L were changed to 0 or 33 mmol/L to keep within the upper and lower bounds of the surveillance error grid.

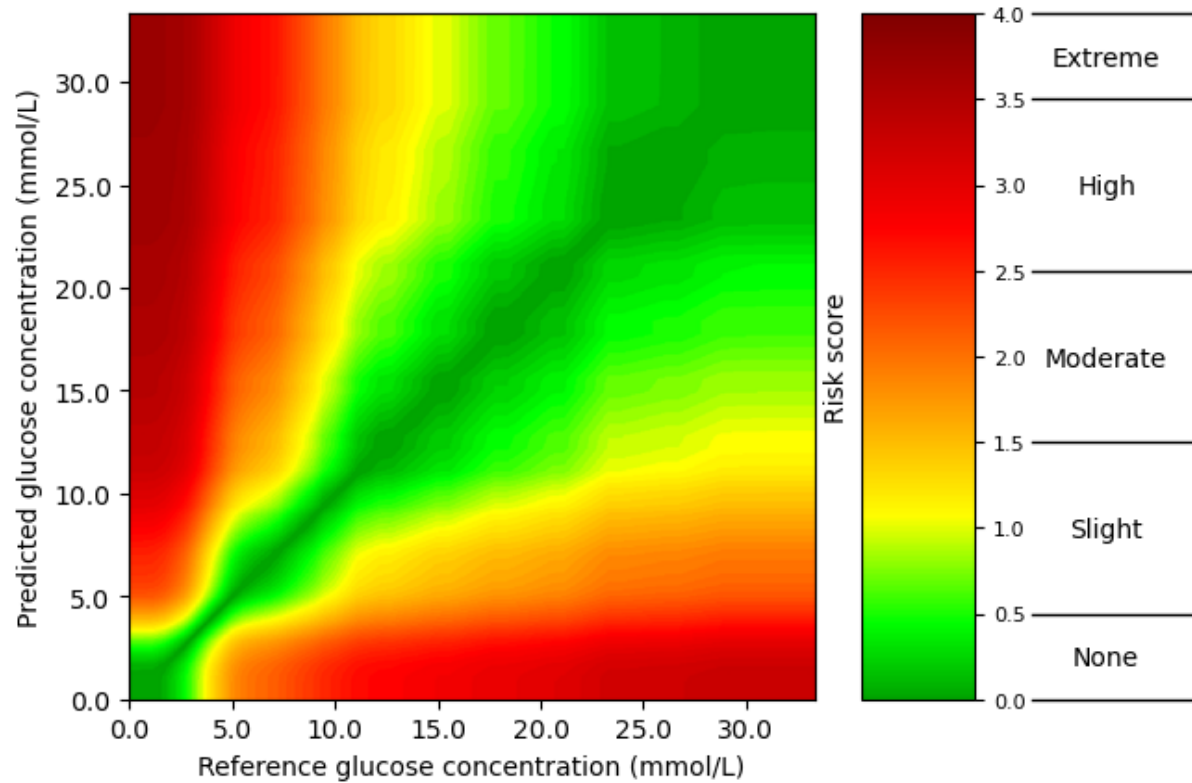


Figure 3.9.: An empty surveillance error grid

3.7. Bias in machine learning models (Study III)

- ☐ Fairness metrics
- ☐ Mitigation strategies -> Fine-tuning LSTM
- ☐ Regression analysis and coefficients?
- ☐ Evaluate bias when output is not binary (e.g., LMEM models)
- ☐ CGM specific error grids

4 Helenes Notes**

- ☒ Study I
- ☐ Study II
- ☐ NT-proBNP as a biomarker of cardiovascular risk
- ☐ Present CGM-derived metrics
- ☒ Study II

5.1. metrics used in the papers - Study 1 - Scoping review

- ☐ CGM metrics and CVD definitions'
- ☐ Demographics
- ☐ Study designs / cohorts
- ☐ Diabetes types
- ☐ CGM metrics
- ☐ CVD risk factors
- ☐ Statistical methods used
- ☐ Check big and small letters in headers (uppercase/lowercase beginnings)

5.2. Study I: Results from a Scoping Review

5.2.1. Study Selection

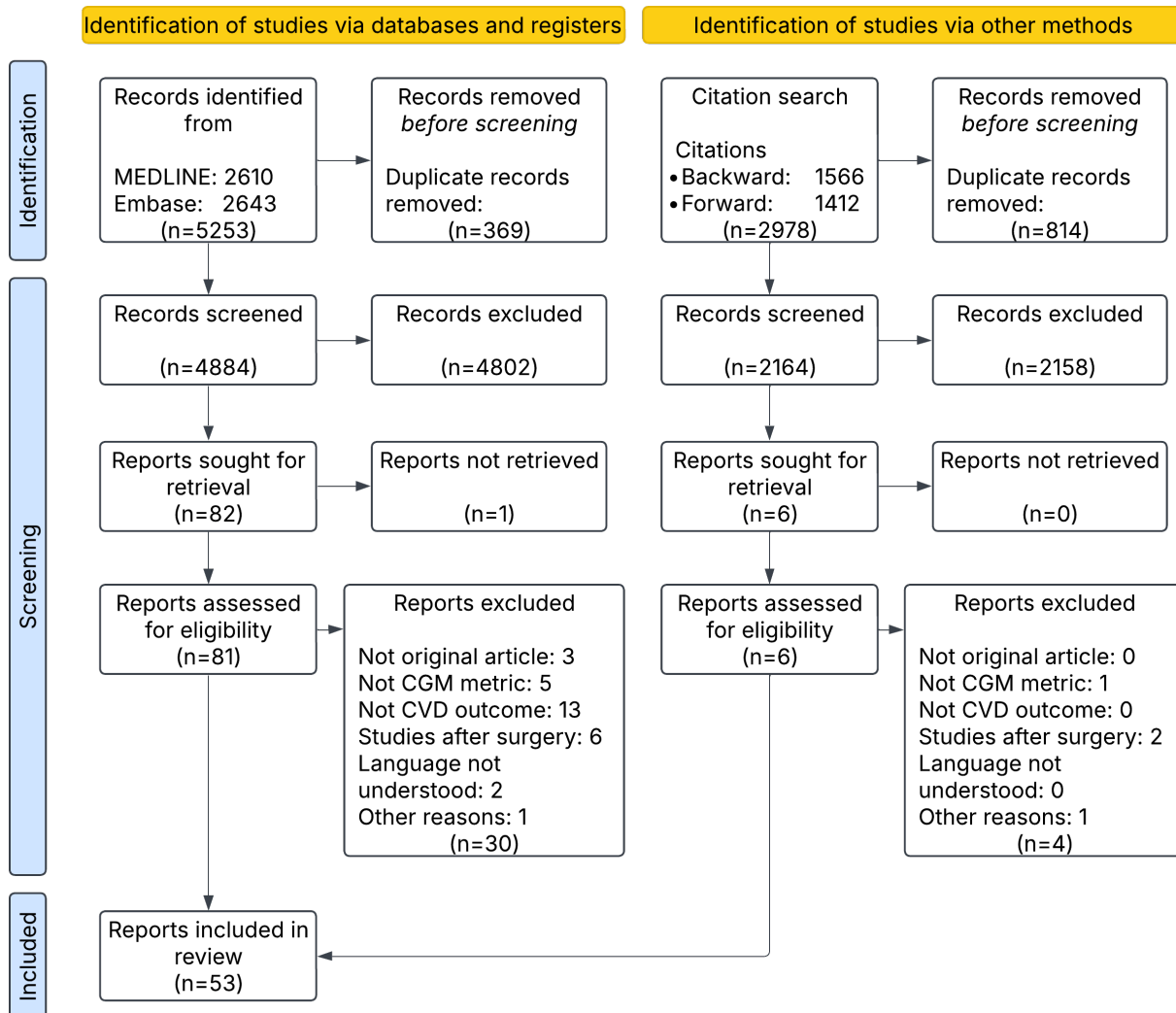
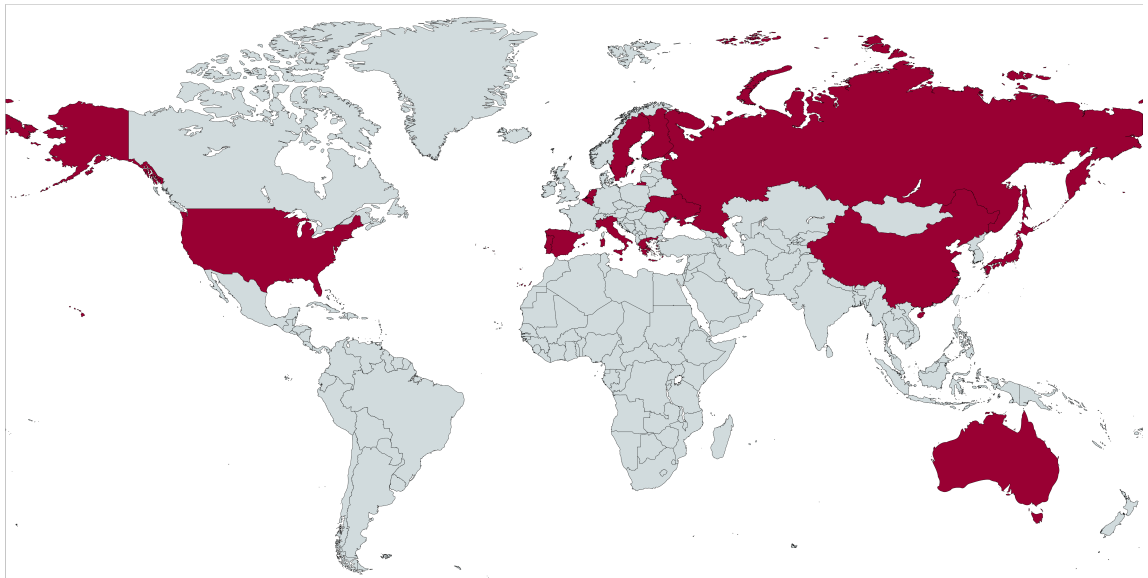


Figure 5.1.: Flow diagram of study selection

5.2.2. Study Populations

- Add mapchart.net to the refernecelist and cite it in this figure text
- If timer permits it then change colorscale to viridis or AU blue ::: {#fig-prisma}



Geographical location of the studies (colored in red) were found based on the hospitals and centers where data was collected or by the first author's affiliation when the other option was not available. Illustration was created with mapchart.net

Geographic distribution of the studies :::

5.2.3. Study Designs and Demographics

- ☐ Check table is correct with the published article
- ☐ Add a raised letter to each abbreviation and include it in the table when relevant
- ☐ Add references to all citations

Table 5.1.: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Chen, 2020 [†]	Longitudinal (prospective) and cross-sectional (retrospective)FU: in-hospital or within 3 months after discharge from hospital	T2D with CAD (Only male)BG control, n = 90BG fluctuation, n = 120	210	BG controls: 55.53 ± 7.30 BG fluctuation: 56.41 ± 7.67	BG controls: 6.59 ± 2.30 BG fluctuation group: 6.92 ± 2.25	nr	2 days
He, 2023	Longitudinal (prospective)FU: 1 year	T2D with kidney disease on hemodialysisHigh TIR, n = 12Low TIR, n = 15	27	High TIR: 66 [63, 73]Low TIR: 70 [64, 75]	High TIR: 2 [1.7, 10]Low TIR: 5 [0.75, 20]	High TIR: 43 [37, 51] mmol/mol (6.1 [5.5, 6.8]%)Low TIR: 66 [46, 70] mmol/mol (8.2 [6.4, 8.6]%)	14 days
Lu, 2021	Longitudinal (prospective)FU: until death occurred or 3–13 yearsMedian: 6.9 years	T2D, hospitalized	6225	mean age = 61.7 years	mean: 9.7 years	74.0 ± 24.0 mmol/mol (8.9 $\pm$ 2.2%)	72 hours

Table 5.1.: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Wei, 2019	Longitudinal (prospective) Median FU [IQR]: 31 months (22, 56)	T2D Divided into three groups: 1) No hypo, n = 11732) Mild hypo (lv11), n = 3233) Severe hypo (lv13), n = 24	1520	No hypoglycemia: 58.59 ± 11.26 Hypoglycemia: 62.27 ± 11.58	No hypoglycemia: 6.46 ± 6.00 Hypoglycemia: 7.78 ± 7.37	No hypoglycemia: $(8.19 \pm 2.10\%)$ Hypoglycemia: $(7.73 \pm 1.96\%)$	3 days
Bezerra, 2023	Cross-sectional	T1D	161	37.4 ± 13.4	17.7 ± 10.6	$(7.5 \pm 1.1\%)$	14 days
De Meulemeester, 2024 [†]	Cross-sectional	T1D	808	44.8 ± 15.2	23.1 ± 13.6	63 ± 13 mmol/mol ($7.9 \pm 1.2\%$)	2 weeks
Deng, 2023	Cross-sectional	T2D	860	HP 1 carriers: 53.5 ± 13.3 HP 2-2: 51.7 ± 14.6	HP 1 carriers: 8.6 ± 6.4 HP 2-2: 8.6 ± 6.7	HP1 carriers: 73.0 ± 24.0 mmol/mol ($8.8 \pm 2.2\%$) HP2-2: 72.0 ± 23.0 mmol/mol ($8.7 \pm 2.1\%$)	3 days
El Malahi, 2022	Cross-sectional	T1D starting on sensor-augmented pump therapy	515	Mean $\pm$ SD [years]: 42.2 ± 12.5	22.3 ± 11.6	60 ± 9.8 mmol/mol ($7.6 \pm 0.9\%$)	2 weeks
Guo, 2021	Cross-sectional	T1D or T2D with atrial fibrillation With Stroke, n = 48 Without Stroke, n = 462	510	Stroke: 70.3 ± 12.1 No stroke: 68.1 ± 9.4	nr	No Stroke: 7.4 ± 2.1 Stroke: 8.2 ± 1.7	72 hour

Table 5.1.: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Li, 2020	Cross-sectional	T2D with LEAD, n = 179 T2D without LEAD, n = 157	336	Without LEAD: 55.94 ± 12.45 With LEAD: 65.56 ± 11.99	Without LEAD: 6.92 ± 3.54 With LEAD: 10.32 ± 4.14	Without LEAD: 7.85 ± 1.41% With LEAD: 8.97 ± 1.63% median: 45 mmol/mol (6.8%)	72 hours
Magri, 2018 [†]	Cross-sectional	T2D	121	64 [57, 68]	3 [2, 5]	Non-CAD: 7.51 ± 0.80% CAD: 7.75 ± 0.92%	72 hours
Mi, 2012	Cross-sectional	T2D with chest pain Without CAD, n = 202 With CAD, n = 84	286	Without CAD: 62.8 ± 8.7 With CAD: 66.6 ± 9.2	nr		72 hours (Only intermediate 48 hours were used)
Sheng, 2023	Cross-sectional	T2D, hospitalized	545	mean age of (61.22 ± 11.21) years	nr	8.51 ± 1.85%	7-14 days
Su, 2011	Cross-sectional	T2D with chest pain Without CAD, n = 92 With CAD, n = 252	344	Without CAD: 61 ± 9 With CAD: 65 ± 9	Without CAD: 4.8 ± 5.7 With CAD: 6.5 ± 6.4	non-CAD: 7.5 ± 1.4% CAD: 7.6 ± 1.5%	72 hours (only 48 hours used)
Watanabe, 2017	Cross-sectional	Prediabetes, hospitalized	28	64.3 ± 12.8	nr	5.41 ± 0.35%	72 hours (Only used the middle 48 hours)

Table 5.1.: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Zhang, 2013	Cross-sectional	T2D with cardiovascular complications- Group A: healthy individuals Group B: T2D without cardiovascular complications- Group C: T2D with cardiovascular complications	92	Group A: 56.3 ± 6.1 Group B: 56.1 ± 6.6 Group C: 61.7 ± 7.2	nr	Group A: $5.3 \pm 0.3\%$ Group B: $6.6 \pm 1.2\%$ Group C: $7.5 \pm 1.4\%$	72 hours

[†] This study appears both in the clinical and subclinical disease outcome tables due to investigating multiple CVD outcomes.

Note: Age is reported in years either as an interval, mean $\pm$ SD, or median (IQR). Diabetes duration values originally reported in months were converted to years for consistency (months $\div$ 12). *Abbreviations:* CGM, continuous glucose monitoring; FU, follow-up; T1D, type 1 diabetes; T2D, type 2 diabetes; CAD, coronary artery disease; hypo, hypoglycemia; LEAD, lower extremity arterial disease; IQR, interquartile range; nr, not reported.

5.2.4. Outcomes from the Studies in the Scoping Review

- ☐ Check table is correct
- ☐ Add references to all citations
- ☐ Correct the raised letters/dagger is the same between the two tables
- ☐ Delete the multimedia part in the footnote of the table

Table 5.2.: Main findings of the included studies on clinical cardiovascular outcomes. {tbl-colwidths='[40, 15, 15, 15, 15]'}

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings
Cardiovascular mortality				
Lu et al [21], 2021				
TIR ^c > 85%	HR 1.00	< .001 (trend)	HR 1.00	.02 (trend)
TIR 71% – 85%	HR 1.43 (0.95–2.14)	< .001 (trend)	HR 1.35 (0.90–2.04)	.02 (trend)
TIR 51% – 70%	HR 1.66 (1.12–2.45)	< .001 (trend)	HR 1.47 (0.99–2.19)	.02 (trend)
TIR ≥ 50%	HR 2.15 (1.47–3.13)	< .001 (trend)	HR 1.85 (1.25–2.72)	.02 (trend)
TIR as a continuous variable (each 10% decrease)	HR 1.08 (1.03–1.13)	— ^d	HR 1.05 (1.00–1.11)	—
Wei et al [46] ^e , 2019				
Hypoglycemic events	HR 2.033 (1.211–3.413)	—	HR 2.642 (1.398–4.994)	—
Major adverse cardiovascular events				
He et al [45], 2023				
Blood glucose risk index	HR 0.97 (0.85–1.10)	.61	HR 0.98 (0.85–1.13)	.75
Low blood glucose index	HR 2.37 (1.16–4.83)	.02	HR 2.73 (1.21–6.16)	.02
High blood glucose index	HR 0.94 (0.81–1.08)	.38	HR 0.94 (0.81–1.09)	.44
Average of daily risk range	HR 1.00 (0.93–1.07)	> .99	HR 1.01 (0.93–1.09)	.80
GMI ^f	HR 0.98 (0.91–1.06)	.65	HR 0.99 (0.91–1.07)	.78
M-value	HR 0.98 (0.91–1.05)	.54	HR 0.98 (0.91–1.06)	.64
Wei et al [46] ^e , 2019				
Hypoglycemic events	HR 1.501 (1.207–1.866)	—	HR 1.615 (1.239–2.106)	< .001
Macrovascular complications				
De Meulemeester et al [48] ^{e,g} , 2024				
TIR	OR 0.939 (0.829–1.063)	> .05	OR 0.896 (0.738–1.087)	> .05
TITR ^h	OR 0.901 (0.775–1.047)	> .05	OR 0.933 (0.745–1.169)	> .05

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings
Bezerra et al [47], 2023				
TIR	OR 0.66 (0.46–0.93)	.02	OR 0.68 (0.39–1.16)	.15
Time below 54 mg/dL	OR 1.10 (0.88–1.38)	.39	OR 0.92 (0.62–1.34)	.65
TBR ⁱ	OR 0.93 (0.80–1.09)	.39	OR 0.77 (0.54–1.11)	.17
TAR ^j	OR 1.04 (1.01–1.08)	.01	OR 1.04 (0.99–1.10)	.08
Time above 250 mg/dL	OR 1.04 (1.00–1.08)	.03	OR 1.03 (0.97–1.09)	.29
CV ^k	OR 1.03 (0.95–1.11)	.52	OR 0.92 (0.81–1.06)	.25
GMI	OR 2.17 (1.14–4.11)	.02	OR 2.03 (0.77–5.37)	.15
Deng et al [49], 2023				
%CV tertile 1 (Hp1; reference)	OR 1.000	.07 (interaction)	OR 1.000	.008 (interaction)
%CV tertile 1 (Hp2-2; reference)	OR 1.000	.07 (interaction)	OR 1.000	.008 (interaction)
%CV tertile 2 (Hp1)	OR 1.483 (0.907–2.423)	.12	OR 1.048 (0.528–2.078)	.89
%CV tertile 2 (Hp2-2)	OR 1.399 (0.829–2.358)	.21	OR 0.659 (0.296–1.466)	.31
%CV tertile 3 (Hp1)	OR 2.347 (1.393–3.957)	.001	OR 2.461 (1.183–5.121)	.02
%CV tertile 3 (Hp2-2)	OR 1.217 (0.731–2.027)	.45	OR 0.540 (0.245–1.191)	.13
El Malahi et al [50], 2022				
TIR	—	—	—	> .05
SD	—	—	—	> .05
CV	—	—	—	> .05
Magri et al [27] ^g , 2018				
TBR	—	—	OR 1.12 (1.014–1.228)	.02
Lowest BG ^l value	—	—	—	—
Area under the TBR curve	—	—	—	—
Coronary artery disease				
De Meulemeester et al [48] ^{e,g} , 2024				
TITR	OR 1.039 (0.812–1.330)	> .05	OR 1.255 (0.874–1.803)	> .05

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings
TIR	OR 1.072 (0.866–1.328)	> .05	OR 1.164 (0.844–1.607)	> .05
Sheng et al [53], 2023				
TIR < 20%	—	—	OR 2.143 (1.554–3.287)	—
TIR 20% – 40%	—	—	OR 1.049 (0.945–2.022)	—
TIR 40% – 60%	—	—	OR 0.854 (0.495–1.473)	—
TIR 60% – 80%	—	—	OR 0.617 (0.423–1.312)	—
TIR > 80%	—	—	OR 0.470 (0.143–1.545)	—
Chen et al [40] ^{e,g} , 2020				
Controls with SD < 1.40 mmol/L, MAGE ^m < 3.90 mmol/L, LAGE ⁿ < 4.40 mmol/L, MODD ^o < 0.83 mmol/L versus high BG fluctuations (myocardial Infarction)	—	—	$\chi^2 = 5.797$	.02
Controls with SD < 1.40 mmol/L, MAGE < 3.90 mmol/L, LAGE < 4.40 mmol/L, MODD < 0.83 mmol/L versus high BG fluctuations (angina pectoris)	—	—	$\chi^2 = 7.490$	.006
Wei et al [46] ^e , 2019				
Hypoglycemic events (myocardial Infarction)	HR 1.901 (1.067–3.389)	—	HR 1.549 (0.768–3.124)	.03
Hypoglycemic events (unstable angina pectoris)	HR 1.226 (0.857–1.753)	—	HR 1.218 (0.794–1.869)	.30
Shu-Hua et al [43] ^e , 2012				
MAGE level (≥ 3.4 mmol/L)	—	—	OR 2.286 (1.176–4.446)	.02

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings
Su et al [28] ^e , 2011				
MAGE ≥ 3.4 mmol/L	—	—	OR 2.612 (1.423–4.831)	.002
MAGE	—	—	AUC 0.618 (0.555–0.680)	.001
Gensini score				
Chen et al [40] ^{e,g} , 2020				
Controls with SD < 1.40 mmol/L, MAGE < 3.90 mmol/L, LAGE < 4.40 mmol/L, MODD < 0.83 mmol/L versus high BG fluctuations	—	—	$t = 6.210$	< .001
Watanabe et al [54] ^e , 2017				
MAGE	—	—	$r = 0.742$	< .001
Shu-Hua et al [43] ^e , 2012				
MAGE	—	—	Unstandardized coefficient $\beta = 4.817$; SE = 1.164; standardized coefficient $\beta = 0.170$; $t = 2.984$	.003
Su et al [28] ^e , 2011				
MAGE	—	—	Unstandardized $\beta = 7.010$; SE = 1.466; standardized $\beta = 0.237$; $t = 4.783$	< .001
Syntax score				
Watanabe et al [54] ^e , 2017				
MAGE	—	—	$r = 0.776$	< .001
Zhang et al [30] ^e , 2013				
MAGE	—	—	$r = 0.518$	.01
BG fluctuations from 00:00 to 03:00	—	—	$r = -0.442$	.04

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings
BG fluctuations from 03:00 to 06:00	—	—	$r = -0.208$	.34
BG fluctuations from 06:00 to 08:00	—	—	$r = 0.678$	< .001
BG fluctuations from 08:00 to 11:00	—	—	$r = 0.115$	.60
BG fluctuations from 11:00 to 13:00	—	—	$r = 0.523$	.01
BG fluctuations from 13:00 to 17:00	—	—	$r = 0.257$	.24
BG fluctuations from 17:00 to 19:00	—	—	$r = 0.358$	.09
BG fluctuations from 19:00 to 24:00	—	—	$r = -0.018$	.93
Stroke				
De Meulemeester et al [48] ^{e,g} , 2024				
TITR	OR 0.651 (0.470–0.902)	< .05	OR 0.546 (0.347–0.858)	< .01
TIR	OR 0.749 (0.588–0.955)	< .05	OR 0.617 (0.440–0.866)	< .01
Guo et al [51], 2021				
TIR: Q1 ($\leq 46\%$; reference)	OR 1.00	< .001	OR 1.00	< .001
TIR: Q2 ($46\% - 65\%$)	OR 0.86 (0.72–0.95)	< .001	OR 0.80 (0.68–0.92)	< .001
TIR: Q3 ($65\% - 81\%$)	OR 0.71 (0.61–0.81)	< .001	OR 0.64 (0.53–0.79)	< .001
TIR: Q4 ($> 81\%$)	OR 0.66 (0.58–0.80)	< .001	OR 0.59 (0.50–0.74)	< .001
TIR (per 10% increase)	OR 0.93 (0.85–0.98)	.008	OR 0.89 (0.82–0.95)	.001
Wei et al [46] ^e , 2019				
Hypoglycemic events	HR 1.691 (1.144–2.499)	—	HR 1.813 (1.110–2.960)	.06
Peripheral artery disease				

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings
De Meulemeester et al [48] ^{e,g} , 2024				
TITR	OR 0.680 (0.426–1.085)	> .05	OR 0.807 (0.382–1.703)	> .05
TIR	OR 0.736 (0.520–1.042)	> .05	OR 0.811 (0.470–1.398)	> .05
Lower extremity arterial disease				
Li et al [52], 2020				
TIR	OR 0.979 (0.968–0.991)	< .001	OR 0.979 (0.965–0.992)	.002
CV	OR 1.040 (1.003–1.078)	.04	OR 1.038 (0.996–1.081)	.08
SD	OR 1.325 (1.038–1.691)	.02	OR 1.158 (0.824–1.627)	.40
TIR-without LEAD ^p (1)	OR 1.00	—	OR 1.00	—
TIR-mild LEAD (1)	OR 0.98 (0.97–1.00)	.14	OR 0.99 (0.97–1.01)	.25
TIR-moderate LEAD (1)	OR 0.97 (0.95–0.99)	.007	OR 0.97 (0.95–0.99)	.01
TIR-without severe LEAD (1)	OR 0.96 (0.94–0.98)	.002	OR 0.96 (0.94–0.98)	.003
CV-without LEAD	—	—	OR 1.00	—
CV-mild LEAD	—	—	OR 1.03 (0.98–1.07)	.28
CV-moderate LEAD	—	—	OR 1.02 (0.96–1.09)	.48
CV-without severe LEAD	—	—	OR 1.02 (0.95–1.09)	.60
TIR-without LEAD (2)	—	—	OR 1.00	—
TIR-mild LEAD (2)	—	—	OR 0.97 (0.96–1.08)	.06
TIR-moderate LEAD (2)	—	—	OR 0.98 (0.95–0.99)	.01
TIR-without severe LEAD (2)	—	—	OR 0.97 (0.95–0.99)	.02
SD-without LEAD	—	—	OR 1.00	—
SD-mild LEAD	—	—	OR 0.88 (0.47–1.64)	.69
SD-moderate LEAD	—	—	OR 1.28 (0.58–3.07)	.58
SD-without severe LEAD	—	—	OR 1.52 (0.92–2.41)	.10

^a The full table with adjustments is provided in **Multimedia Appendix 10**. Further elaboration on the adjusted variables can be found in **Multimedia Appendices 7 and 8**.

^b All hazard ratios (HRs), odds ratios (ORs), and areas under the curve (AUCs) are reported as follows: point estimate (95% CI).

^c TIR: time in range.

^d Not applicable or not available/not reported.

^e This study appears multiple times as it investigated multiple cardiovascular disease outcomes.

^f GMI: glucose management indicator.

^g This study appears in both the clinical and subclinical disease outcome tables owing to the investigation of multiple cardiovascular disease outcomes.

^h TITR: time in tight range.

ⁱ TBR: time below range.

^j TAR: time above range.

^k CV: coefficient of variation.

^l BG: blood glucose.

^m MAGE: mean amplitude of glycemic excursions.

ⁿ LAGE: largest amplitude of glycemic excursions.

^o MODD: mean of daily differences.

^p LEAD: lower extremity arterial disease.

Table 5.3.: Summary of established clinical and subclinical outcomes and corresponding abbreviations identified in the literature.

Clinical	Subclinical
• CVD = Cardiovascular disease - Macrovascular complications - Diabetic macroangiopathy • CAD = Coronary artery disease - CHD = Coronary heart disease - MI = Myocardial infarction - Heart attack - Heart failure - (Unstable) angina pectoris • Cardiovascular events - Ischemic heart disease (IHD) • LEAD = Lower extremity artery disease • MACE = Major adverse cardiovascular events • PAD = Peripheral artery disease • Stroke = Stroke, Cerebrovascular disease, or Cerebrovascular accident	• ABI = Ankle-brachial index • BP = Blood pressure • Carotid artery distensibility = Carotid artery distensibility or carotid distensibility coefficient • CCA = Common carotid artery • CIMT = Carotid artery intima-media thickness • DBP = Diastolic blood pressure • FMD = Flow-mediated dilation of the brachial artery • GSM = Gray-scale median of the carotid arteries • HRV = Heart rate variability • IMT = Intima-media thickness • PWV = Pulse wave velocity - ba-PWV = Brachial-ankle pulse wave velocity - cr-PWV = Carotid-radial pulse wave velocity - cf-PWV = Carotid-femoral pulse wave velocity - Aortic PWV = Aortic pulse wave velocity • SBP = Systolic blood pressure

5.2.5.

5.3. Baseline characteristics table of the datasets

Table 5.4.: Baseline table for CANCAN data and NT-proBNP groups.

Variable	CANCAN, total	NT-proBNP 125 pg/mL	NT-proBNP > 125 pg/mL
N	159	92	67
Age (years)	65 [56, 71]	60 [53, 68]	68 [64, 76]
Sex			
– Male	99 (62%)	55 (60%)	44 (66%)
– Female	60 (38%)	37 (40%)	23 (34%)
Smoking status			
– Never or former	135 (85%)	77 (84%)	58 (87%)
– Current	24 (15%)	15 (16%)	9 (13%)
Diabetes duration (years)	16 [11, 24]	14 [9, 22]	19 [14, 25]
SCORE2-Diabetes (%)	14.8 [9.1, 20.9]	11.9 [7.8, 17.2]	19.0 [13.4, 25.4]
HbA1c (mmol/mol)	64 [55, 80]	66 [56, 86]	60 [54, 72]
Systolic blood pressure (mmHg)	132 [123, 143]	132 [123, 143]	131 [122, 142]
HDL (mmol/L)	1.0 [0.9, 1.2]	1.0 [0.9, 1.2]	1.0 [0.9, 1.3]
Total cholesterol (mmol/L)	3.9 [3.2, 4.8]	4.1 [3.5, 4.9]	3.6 [3.0, 4.2]
eGFR (mL/min/1.73 m ²)	57 [39, 74]	68 [52, 82]	48 [27, 63]
CGM-derived metrics			
– CGM days (n/participant)	13 [10, 13]	13 [9, 13]	13 [10, 13]

Variable	CANCAN, total	NT-proBNP 125 pg/mL	NT-proBNP > 125 pg/mL
– Glucose management indicator (%)	7.3 [6.8, 8.4]	7.5 [6.8, 8.6]	7.2 [6.7, 8.0]
– Time in range (%)	59.6 [31.9, 78.9]	58.2 [20.2, 78.6]	62.1 [40.8, 81.0]
– Time below range (%)	0.0 [0.0, 1.5]	0.0 [0.0, 0.8]	0.2 [0.0, 2.0]
– Time above range (%)	36.6 [17.6, 68.1]	41.8 [19.8, 79.5]	33.1 [14.1, 58.2]
– Coefficient of variation (%)	27.3 [22.5, 31.9]	25.3 [21.2, 30.3]	28.3 [24.8, 34.2]
– Mean amplitude of glycemic excursions (mmol/L)	6.3 [5.0, 7.9]	6.0 [4.9, 7.7]	6.7 [5.1, 8.5]

Categorical values are given by number (percentage), and numerical values are given by median [interquartile range].

Table 5.5.: Baseline table for T1DX data and racial groups.

Characteristics	Total	White Cohort	Black Cohort
N	205	101	104
Race (White)	101 (49%)	101 (100%)	0 (0%)
Sex (male)	88 (43%)	46 (46%)	42 (40%)
Age group (child, < 18 yrs)	82 (40%)	49 (49%)	33 (32%)
Age (years)	28 [15, 47]	22 [13, 46]	32 [16, 47]
BMI (kg/m ²)	25.1 [21.7, 29.2]	24.0 [20.7, 28.6]	26.7 [22.3, 30.0]
Age at T1D diagnosis	11 [7, 22]	9 [6, 16]	14 [8, 25]
HbA1c (mmol/mol)	67 [57, 78]	63 [55, 73]	72.0 [60, 85]
CGM duration w/o missing data (weeks/participant)	9.1 [7.2, 10.3]	9.5 [7.7, 10.7]	8.6 [6.9, 9.9]
Missing CGM values (weeks/per participant)	3.3 [1.6, 5.1]	2.6 [1.3, 4.5]	4.1 [1.8, 5.6]
CGM-derived metrics			
– CGM days (n/participant)	58 [46, 69]	63 [47, 71]	56 [45, 66]
– Glucose management indicator (%)	7.5 [7.0, 8.2]	7.4 [6.9, 7.9]	7.7 [7.0, 8.5]
– Time in range (%/per participant)	47 [35, 57]	49 [39, 60]	42 [31, 54]
– Time above range (%/per participant)	45 [30, 56]	42 [28, 53]	48 [32, 60]
– Time below range (%/per participant)	8 [4, 14]	6 [4, 10]	11 [5, 15]
– Coefficient of variation (%)	43.3 [39.6, 48.6]	42.2 [38.6, 46.5]	46.7 [40.4, 51.9]

Characteristics	Total	White Cohort	Black Cohort
– Mean amplitude of glycemic excursions (mmol/L)	9.9 [8.2, 12.1]	9.6 [8.1, 11.2]	10.6 [8.5, 13.0]

Categorical values are given by number (percentage), and numerical values are given by median [interquartile range].

5.4. Study II: Results from a Foundation Model Embedding Analysis

5.4.1. Representation learning of CGM data (Study 2)

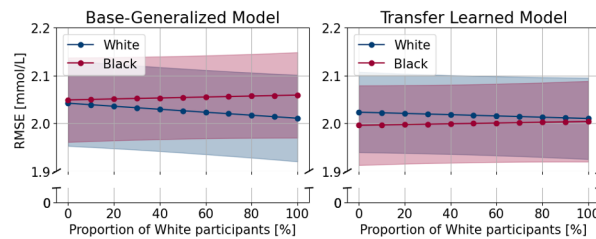
5.4.2. Prediction of NT-proBNP (Study 2)

- Prediction NT-proBNP with different baseline models
- Predicting NT-proBNP using only age
- Predicting NT-proBNP using only CGM metrics
- Predicting NT-proBNP using SCORE2-Diabetes inputs and score together with other CGM data -> Supplementary material of paper 2

5.5. Study III: Results from an Algorithmic Fairness Study

5.5.1. Bias in Machine Learning models (Study 3)

- Baseline characteristics of the T1DX cohort stratified by race status
- Fairness metrics of the LMEM model before and after fine-tuning



Note: The y-axis starts at 1.9 mmol/L and not 0

Figure 5.2.: Estimated root mean squared errors (RMSE) across 11 different proportions of White and Black participants in the dataset.

Ratio	LOCF	Base-Individual	Base-Generalized	Transfer Learned
Model performance (RMSE) for White participants				
0			2.04 [1.95, 2.13]	2.02 [1.94, 2.11]
10			2.04 [1.95, 2.13]	2.02 [1.94, 2.11]
20			2.04 [1.95, 2.12]	2.02 [1.94, 2.10]

Ratio	LOCF	Base-Individual	Base-Generalized	Transfer Learned
30			2.03 [1.94, 2.12]	2.02 [1.94, 2.10]
40	2.45 [2.32, 2.57]	2.54 [2.38, 2.70]	2.03 [1.94, 2.12]	2.02 [1.94, 2.10]
50	(independent of ratio)	(independent of ratio)	2.03 [1.94, 2.11]	2.02 [1.93, 2.10]
60			2.02 [1.94, 2.11]	2.02 [1.93, 2.10]
70			2.02 [1.93, 2.11]	2.01 [1.93, 2.10]
80			2.02 [1.93, 2.11]	2.01 [1.93, 2.10]
90			2.01 [1.92, 2.10]	2.01 [1.93, 2.10]
100			2.01 [1.92, 2.10]	2.01 [1.92, 2.10]
Model performance (RMSE) for Black participants				
0			2.05 [1.96, 2.14]	2.00 [1.91, 2.08]
10			2.05 [1.96, 2.14]	2.00 [1.91, 2.08]
20			2.05 [1.96, 2.14]	2.00 [1.92, 2.08]
30			2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
40	2.41 [2.28, 2.53]	2.63 [2.47, 2.79]	2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
50	(independent of ratio)	(independent of ratio)	2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
60			2.06 [1.97, 2.14]	2.00 [1.92, 2.08]
70			2.06 [1.97, 2.14]	2.00 [1.92, 2.08]
80			2.06 [1.97, 2.14]	2.00 [1.92, 2.09]
90			2.06 [1.97, 2.15]	2.00 [1.92, 2.09]
100			2.06 [1.97, 2.15]	2.00 [1.92, 2.09]

Values that vary by proportion are estimated using linear mixed-effects models. These models were adjusted to the population's age (40% children) and sex (57% women) distribution. Abbreviations: LOCF = last observation carried forward.

Root mean squared errors with 95% CIs for the different models.

Predictor	Coefficient	95% CI	P-value
Intercept	1.893	[1.762, 2.024]	<0.001
Race			
White	Reference		
Black	0.007	[-0.118, 0.132]	0.91
Age			
Adult	Reference		
Child	0.482	[0.355, 0.610]	<0.001
Sex			
Male	Reference		
Female	-0.076	[-0.201, 0.048]	0.23
Training size	0.019	[-0.055, 0.093]	0.62
Base-Generalized (per 100,000 samples)*			
Ratio (per 10%)	-0.003	[-0.007, 0.001]	0.12
Ratio · Race	0.004	[0.001, 0.008]	0.02

*Training size was centered at 510,000 as the training data had a mean of 514,293 samples.

5. Results

Model coefficients from linear mixed effects models for root mean squared errors of the Base-Generalized models.

Lorum ipsum Lorum ipsum Lorum ipsum Lorum ipsum Lorum ipsum Lorum ipsum Lorum ipsum Lorum ipsum Lorum ipsum

Predictor	Coefficient	95% CI	P-value
Intercept	1.890	[1.768, 2.015]	<0.001
Race			
White	Reference		
Black	-0.028	[-0.147, 0.092]	0.65
Age			
Adult	Reference		
Child	0.411	[0.290, 0.533]	<0.001
Sex			
Male	Reference		
Female	-0.057	[-0.173, 0.060]	0.34
Training size	0.011	[-0.060, 0.081]	0.77
Base-Generalized (per 100,000 samples)*			
Training size tl (per 1,000 samples)**	-0.111	[-0.161, -0.061]	<0.001
Ratio (per 10 units)	-0.001	[-0.003, 0.001]	0.46
Ratio · Race	0.002	[-0.001, 0.005]	0.20

*Training size was centered at 510,000 as the training data had a mean of 514,293 samples.

**Training size for the transfer learned approach was centered at 4,400 as the training data for fine-tuning had a mean of 4,371 samples.

Model coefficients from linear mixed effects models for root mean squared errors of the Transfer Learned models.

Discussion 6

7 Main findings

- Scoping review highlighted a chaotic landscape of wildly different study designs in the literature
- Mention other reviews, Yapanis + the other one that I forgot
- Representation learning of CGM data can capture complex patterns in glucose dynamics that are not captured by traditional CGM metrics
- Machine learning models can predict NT-proBNP levels using CGM data, however the performance of these models is modest and there is still room for improvement
- Bias is a concern in machine learning models, and fine-tuning on a subset of the data can help mitigate bias as measured by fairness metrics
- Grouped fairness is easier to evaluate and mitigate than individual fairness, but it is still an important step in ensuring that machine learning models are equitable and do not perpetuate existing biases in healthcare.

Machine learning vs traditional statistical methods 8

9 something about data and cross-sectional v

- Mention the collins protocol that creates a study focused on answer this question
- Mention the CANCAN dataset and the weird trend with people being better of the more comobidities
- Why not use CONGA; AI: In participants with insufficient valid CGM observations, CONGA could not be calculated reliably using the available implementation

9.1. Study 3 Method choices

PREDICTION HORIZON Many studies in the literature tests a variety of different prediction horizons but it seems to always predict best, the shorter the prediction horizon is. Most often sensors do also have a higher resolution then 15 minuts (3-5 minutes resolution)

- ☐ In the discussion I could mention LSTM vs CNN vs MLP and talk about the difference in model doesn't make that big of a difference.

Clinical implications 10

11 Strengths and limitations

Conclusions and future perspectives 12

13 Conclusion

- CGM data has the potential to provide valuable insights into cardiovascular risk in diabetes, but there are still many challenges to overcome in terms of study design, data quality, and machine learning methods.

Perspectives 14

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English summary

Background: Type 2 diabetes (T2D) is an increasing burden on healthcare services due to changes in demography and lifestyle. Migrants are particularly vulnerable to T2D, having a higher prevalence of the disease compared to native populations in western Europe, with some evidence suggesting a higher risk of complications and mortality. Evidence on disparities in care that may contribute to this excess risk in migrants is sparse, but register-based studies are well-suited to explore this in a Danish context. However, these studies are vulnerable to biases due to the lack of a validated method to identify T2D cases in the general population.

Aims: This dissertation aimed to develop and validate a register-based classification of type 1 diabetes (T1D) and T2D, and utilise it to explore specific areas of monitoring, biomarker levels, and pharmacological treatment, where disparities in T2D care between migrant groups and native Danes may contribute to excess complication risk in migrants.

Methods: Three cross-sectional register-based studies were performed. Study I validated two register-based classifications of diabetes - the *Open-Source Diabetes Classifier* (OSDC), and the *Register of Selected Chronic Diseases* (RSCD) - in a population of survey respondents from Central Denmark Region (*Health In Central Denmark*), while the two subsequent studies examined T2D prevalence and care in nationwide populations, using native Danes as the reference group. Study II computed relative risk (RR) of prevalent T2D and non-fulfilment of eleven indicators corresponding to guideline-recommendations. Study III computed RR of using glucose-lowering drug (GLD) combination therapy, and RR of using each type of GLD.

Results: In study I, both register-based diabetes classifiers identified valid populations of T1D and T2D in a general population (positive predictive values from 87.5% to 94.4%, negative predictive values above 98.9%), although sensitivity in the OSDC was substantially higher compared to the RSCD in both T1D (77.3% vs. 70.0%) and in T2D (94.4% vs. 90.5%). Neither algorithm was able to accurately classify diabetes type in individuals with T1D onset after age 40, nor T2D onset before age 40. In study II, prevalence of T2D was higher in all migrants excluding the Europe-group (fully-adjusted RR from 0.96 [95% confidence interval: 0.94-0.99] (Europe group) to 4.04 [3.89-4.20] (Sri Lanka group)). Non-fulfilment of recommendations for care was common regardless of origin: in eight of the eleven indicators of care studied >25% of native Danes did not meet guideline-recommendations. Apart from monitoring in the Sri Lanka group, migrants were at similar or higher risk of non-fulfilment than native Danes across all indicators of monitoring (crude RR from 0.64 [0.51-0.80] (hemoglobin-A1c (HbA1c) monitoring, Sri Lanka group) to 1.62 [1.36-1.95] (HbA1c monitoring, Somalia group)), HbA1c control (crude RR from 1.27 [1.24-1.30] (Middle East group) to 1.46 [1.41-1.52] (Pakistan group)), and low-density lipoprotein cholesterol (LDL-C) control (crude RR from 1.08 [1.03-1.14] (Former Yugoslavia group) to 1.78 [1.67-1.90] (Somalia group)), while no overall disparities were observed for the presence of pharmacological treatment (crude RR from 0.61 [0.46-0.80] (GLD, Sri Lanka group) to 1.67 [1.34-2.09] (GLD, Somalia group)). In study III, migrants were less likely to use GLD combination therapy than native Danes (fully adjusted RR from 0.77 [0.71-0.85] (Somalia group) to 1.00 [0.97-1.04] (Former Yugoslavia group)). Migrants were more likely to use oral GLD types (fully adjusted RR (use of any oral GLD) from 0.99 [0.97-1.01] (Europe group) to 1.09 [1.06-1.11] (Sri Lanka group)), but less likely to use injection-based GLD types such as insulins

(fully adjusted RR from 0.66 [0.62-0.71] (Sri Lanka group) to 0.94 [0.89-0.99] (Europe group)) and, especially, glucagon-like peptide-1 receptor agonists (GLP1RA) (fully adjusted RR from 0.29 [0.22-0.39] (Somalia group) to 0.95 [0.89-1.01] (Europe group)).

Conclusion and perspectives This dissertation was able to accurately identify populations of T1D and T2D in register data from the general Danish population. Diabetes epidemiologists can rely on the OSDC and the RSCD to provide valid study populations for Danish register-based research, but researchers should be aware of limitations in cases with atypical age at onset and potential challenges due to evolving GLD indications in the future. This dissertation found room for improvement in T2D care in Denmark, and presented several disparities in T2D care between migrants and native Danes. These may contribute excess risk of complication and mortality in migrants, especially among migrants from Somalia, who received the poorest care of all groups overall. Overall, the large disparity in use of GLP1RA between migrants and native Danes may provide a target for future interventions to improve care and reduce complication risk among migrants with T2D.

Dansk resumé

Baggrund: Type 2 diabetes (T2D) er en stigende belastning på sundhedsvæsenet på grund af ændringer i demografi og livsstil. Indvandrere er særligt sårbare overfor T2D, idet de har en højere forekomst af sygdommen i forhold til indfødte befolkninger i Vesteuropa, ligesom nogle undersøgelser peger på en højere risiko for diabetiske komplikationer og død. Potentielle uligheder i behandlingen, der kan bidrage til den øgede risiko hos indvandrere med T2D, er dårligt belyst, men registerbaserede studier er velegnede til at undersøge dette i en dansk sammenhæng. Disse undersøgelser er imidlertid sårbare, idet der ikke foreligger nogen validerede metoder til at identificere T2D-tilfælde i den generelle danske befolkning.

Formål: Denne afhandling havde til formål at udvikle og validere en registerbaseret klassifikation af type 1 diabetes (T1D) og T2D, og anvende den til at undersøge specifikke områder i behandlingen af T2D - kontroller, biomarkør-niveauer og medicinsk behandling - hvor uligheder mellem indvandrergrupper og indfødte danskere kan bidrage til øget risiko for komplikationer hos indvandrere.

Metoder: Som led i afhandlingen blev tre tværsnitsstudier baseret på register-data gennemført. Studie I validerede to registerbaserede klassifikationer af diabetes - *Open-Source Diabetes Classifier* (OSDC) og *Register for Udvalgte Kroniske Sygdomme* (RUKS) - blandt spørgeskema-besvarelser fra personer i Region Midtjylland (*Helbred i Midt*), mens de to efterfølgende studier undersøgte forekomst og behandling af T2D i hele landet og sammenlignede indvandrergrupper med indfødte danskere. Studie II beregnede relativ risiko (RR) for forekomst af T2D og for manglende opfyldelse af elleve indikatorer for behandlingskvalitet svarende til anbefalinger i gældende kliniske retningslinjer. Studie III beregnede RR for brug af kombinationsbehandling med blodsukker-nedsættende lægemidler (GLD) og RR for brug af hver type GLD.

Resultater: I studie I identificerede begge registerbaserede diabetesklassifikationer gyldige populationer af T1D og T2D i en bred befolkning (positive prædiktive værdier fra 87,5% til 94,4%, negative prædiktive værdier over 98,9%), dog med en væsentlig højere sensitivitet i OSDC sammenlignet med RUKS både i T1D (77,3% vs. 70,0%) og i T2D (94,4% vs. 90,5%). Ingen af algoritmerne var i stand til nøjagtigt at klassificere diabetestype hos personer med T1D debut efter 40-årsalderen eller T2D debut før 40-årsalderen. I studie II var forekomsten af T2D højere i alle indvandrergrupper undtagen Europa-gruppen (fuldt justeret RR fra 0,96 [95% konfidensinterval: 0,94-0,99] (Europa-gruppen) til 4,04 [3,89-4,20] (Sri Lanka-gruppen)). Manglende opfyldelse af de vejledende behandlingsanbefalinger var hyppig - uanset oprindelse: i otte ud af elleve indikatorer opfyldte >25% af indfødte danskere ikke anbefalingerne. Fraset rutinemæssig kontrolundersøgelser i Sri Lanka-gruppen, havde indvandrergrupperne tilsvarende eller højere risiko end indfødte danskere for ikke at opfylde behandlingsanbefalinger på tværs af alle indikatorer for rutinemæssig kontrolundersøgelser (rå RR fra 0,64 [0,51-0,80] (hæmoglobin-A1c (HbA1c-undersøgelse, Sri Lanka-gruppen) til 1,62 [1,36-1,95] (HbA1c-undersøgelse, Somalia-gruppen)), blodsukkerniveau (rå RR fra 1,27 [1,24-1,30] (Mellemøsten-gruppen) til 1,46 [1,41-1,52] (Pakistan-gruppen)) og kolesterolniveau (rå RR fra 1,08 [1,03-1,14] (Eksjugoslavien-gruppen) til 1,78 [1,67-1,90] (Somalia-gruppen)), mens der ikke sås nogen gennemgående forskelle for tilstedeværelse af medicinsk behandling (rå RR fra 0,61 [0,46-0,80] (GLD, Sri Lanka-gruppen) til 1,67 [1,34-2,09] (GLD, Somalia-gruppen)). I studie III var indvandrere mindre tilbøjelige til at

få GLD kombinationsbehandling end indfødte danskere (fuldt justeret RR fra 0,77 [0,71-0,85] (Somalia-gruppen) til 1,00 [0,97-1,04] (Eksjugoslavien-gruppen)). Indvandrere var mere tilbøjelige til at benytte orale GLD typer (fuldt justeret RR (brug af ethvert oralt GLD) fra 0,99 [0,97-1,01] (Europa-gruppen) til 1,09 [1,06-1,11] (Sri Lanka-gruppen)), men mindre tilbøjelige til at benytte injektionsbaserede GLD typer så som insulin (fuldt justeret RR fra 0,66 [0,62-0,71] (Sri Lanka-gruppen) til 0,94 [0,89-0,99] (Europa-gruppen)) og især glucagon-like peptide-1 receptor agonister (GLP1RA) (fuldt justeret RR fra 0,29 [0,22-0,39] (Somalia-gruppen) til 0,95 [0,89-1,01] (Europa-gruppen)).

Konklusion og perspektiver: Denne ph.d.-afhandling kunne præcist identificere befolkninger med T1D og T2D i den brede danske befolkning ud fra registerdata. Diabetesforskere kan anvende OSDC og RUKS til at danne gyldige studiepopulationer ud fra de danske registre, men bør være opmærksomme på deres begrænsninger, herunder blandt patienter med atypisk alder for sygdomsdebut samt potentielle udfordringer som følge af udviklingen i brugen af GLD i fremtiden. Afhandlingen fandt plads til forbedringer i behandlingen af T2D i Danmark, og viste flere uligheder mellem indvandrere og indfødte danskere i behandlingen af T2D, som kan medføre øget risiko for komplikationer og død blandt indvandrere - særligt blandt indvandrere fra Somalia, som havde den laveste behandlingskvalitet af alle grupper. Den store forskel i brugen af GLP1RA mellem indvandrere og indfødte danskere kan muligvis være et mål for fremtidige interventioner til at øge behandlingskvaliteten og mindske komplikationsrisikoen blandt indvandrere med T2D.

Search strings (Study I) A

Table A.1.: Search in Ovid MEDLINE (Search date: 22 January 2024)

#	Searches	Results
1	Blood Glucose	10241
2	Self-Monitoring/ Continuous Glucose Monitoring/	34
3	((blood-glucose or bloodglucose or glucose) adj2 (monitor\$ or selfmonitor))orcgmorbgmoriscgmorrtecmor(timeadj3range)).ti, ab, kf, kw or bloodsugar\$ or glucose\$ or glyc?emic\$ or sugar).ti, ab, kf, kw.)adj5(flashorfluctuat or intermittent\$ or scan\$ or variabilit).ti, ab, kf, kw. 8669 5 or/1— 4 38366 6 expCardiovascularDiseases/ 2760764 7 (cardiovasc or cvd or cardiometabol\$ or cardio-metabol\$ or heart\$ or macrovasc\$ or vasc).ti, ab, kw, kf. 2182473 8 or/6— 7 3971342 9 expDiabetesMellitus/ 518864 10 (diabet or dm1 or dm2 or dmi or dmii or iddm or niddm or t1d or t2d).ti,ab,kw,kf. or/9-10	815092
11	or/9-10	875527
12	and/5,8,11	2416
13	exp Animals/ not Humans/	5189493
14	12 not 13 Updated search: 11 March 2025	2300
15	limit 14 to (da=“20231201-20261231” or dt=“20231201-20261231” or ez=“20231201-20261231” or ed=“20231201-20261231”) or/14-15	310
16		2610

Table A.2.: Search in Embase (Ovid) (Search date: 22 January 2024)

#	Searches	Results
1	blood glucose monitoring/	37574
2	exp continuous glucose monitoring system/	5648
3	((blood-glucose or bloodglucose or glucose) adj2 (monitor\$ or selfmonitor))orcgmorbgmorisecgmorrtecmor(timeadj3range)).ti, ab, kf, kw. 4850 or bloodsugar\$ or glucose\$ or glyc?emic\$ or sugar).ti, ab, kf, kw.)adj5(flashorfluctuat or intermittent\$ or scan\$ or variabilit).ti, ab, kf, kw. 16430 5 or/1— 4 81510 6 expcardiovascularisease/ 5122163 7 (cardiovasc or cvd or cardiometabol\$ or cardio-metabol\$ or heart\$ or macrovasc\$ or vasc).ti, ab, kw, kf. 3044925 8 or/6— 7 6303633 9 expdiabetesmellitus/ 1239184 10 (diabet or dm1 or dm2 or dmi or dmii or iddm or niddm or t1d or t2d).ti,ab,kw,kf.	1237019
11	or/9-10	1501812
12	and/5,8,11	11217
13	(rat or rats or mouse or mice or swine or porcine or murine or sheep or lambs or pigs or piglets or rabbit or rabbits or cat or cats or dog or dogs or cattle or bovine or monkey or monkeys or trout or marmoset\$1).ti. and animal experiment/	1237966
14	animal experiment/ not (human experiment/ or human/)	2601109
15	or/13-14	2672240
16	12 not 15	10861
17	limit 16 to exclude medline journals Updated search: 11 March 2025	2176
18	limit 17 to (dc="20231201-20261231" or dd="20231201-20261231" or rd="20231201-20261231")	467
19	or/17-18	2643

Full papers **B**

B.1. Paper I

B.2. Paper II

B.3. Paper III

C Addenda and errata

This chapter contains an overview of:

- Addenda: contents added since the submitted version of the dissertation.
- Errata: corrections to errors discovered since submission of the dissertation.

C.1. Addenda

C.2. Errata